String matching

Find every occurrence of a pattern in a text, including overlaps. Compare direct search, rolling hashes, finite automata and KMP.

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A roadmap for matching

Define the output and edge cases first. Then trace filtering, state transitions and prefix fallback; distinguish correctness from running time.

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String matching

Text T has length n and pattern P has length m. Report every zero-based s with T[s:s+m]=P, including overlapping matches.

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Boundary contract

The empty pattern matches all n+1 boundaries 0..n. A nonempty pattern longer than the text has no matches. The remaining algorithms first assume $1 \le m \le n$.

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Four methods

Naive compares directly. Rabin–Karp filters by hash. A DFA precomputes transitions. KMP uses prefix links to fall back on demand.

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Naive matching

For every s=0..n−m compare P[j] with T[s+j] in order. Report s if all match. Increment s by one to retain overlaps.

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Naive trace

T=ababa, P=aba. At s=0 compare a,b,a and match; s=1 fails at b≠a; s=2 compares a,b,a and matches. Output 0,2.



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Naive cost

At most $(n-m+1)m$ comparisons give worst-case $O((n-m+1)m)$ time and $O(1)$ extra space. Repeated characters can approach the worst case.

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Rabin–Karp fingerprints

For symbols in 0..d−1, hash a length-m string as Σ x[j]d^(m−1−j) mod q. Equal strings have equal hashes; the converse is false.

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Rolling hash

Let h=d^(m−1) mod q. Remove a and append b using t′=(d(t−a h)+b) mod q. Normalize negative remainders into 0..q−1.

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Deriving the rolling update

Let $W=T[s:s+m]$, and let $c=T[s+m]$ be the next symbol. Treat symbols as values and write the unreduced polynomial $H(W)=\Sigma(j=0..m-1)\ W[j]d^{(m-1-j)}$.

Remove the high term $W[0]d^{(m-1)}$, multiply by d, and add c:

$H(T[s+1:s+m+1]) = d(H(W)-W[0]d^{(m-1)})+c$.

Each remaining term shifts one position. Reduce modulo q; with $t_s=H(W) \bmod q$ and $h=d^{(m-1)} \bmod q$, $t_{(s+1)}=(d(t_s-W[0]h)+c) \bmod q$. Any $q \geq 2$ works; primality is unnecessary here.

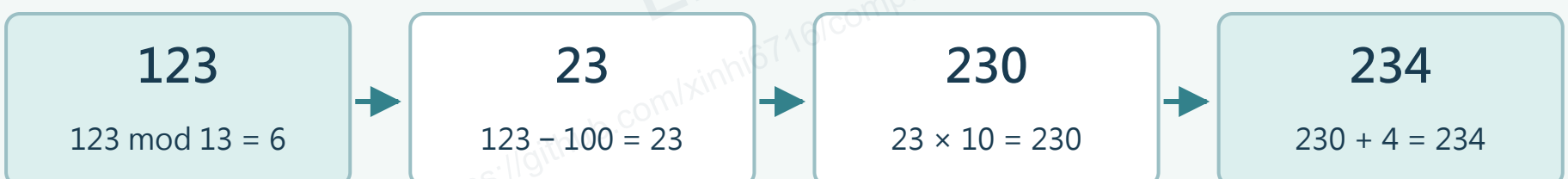
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Rolling-scan loop invariant

Before processing shift s, t_s is the hash of the current window T[s:s+m], and p is the pattern hash. Horner's rule initializes p and t_0. If $s<n-m$, the identity on the preceding slide establishes the invariant for s+1; the last window does not read T[n].

Equal strings have equal hashes, so no true match is filtered out. Verify characters on equal hash, so collisions cannot cause false reports. An intermediate difference may be negative: normalize the final remainder to 0..q−1.

d=10, q=13, m=3: window 123 → 234



h=100 mod 13=9; (10·(6−1·9)+4) mod 13=0.
Normalize negative remainders; 234 mod 13=0.

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Rolling hash trace

With d=10, q=13, m=2, T=1234 and P=34, window hashes for 12,23,34 are 12,10,8. The pattern hash is 8; confirm the final window character by character.

Slide the window one position



remove 1 → ×10 → append 3
(10·(12 − 1·10) + 3) mod 13 = 10

remove 2 → ×10 → append 4
(10·(10 − 2·10) + 4) mod 13 = 8

h = 10; reducing an intermediate value modulo 13 preserves the result.

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Collisions and cost

For $1 \le m \le n$ and constant-time modular arithmetic, let H count all equal-hash windows, including true matches. Verification gives $O(n+m+mH)$, with worst case $\Theta(n+m+m(n-m+1))$.

If $T=a^n$ and $P=a^{\lfloor n/2 \rfloor}$, every window is a true match and verification costs $\Theta(nm)$, even with no false hits.

If a randomized hash gives each unequal window collision probability at most $1/q$, and V is the number of true matches, expected time is $O(n+m+mV+mn/q)$. Near-linear time also requires bounded true-match verification, for example $V=O(1)$ and $q \ge m$.

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Random-base hashing: assumptions

Fix the input first, then choose b uniformly from the finite field F_q for prime q. Distinct symbols have distinct encodings in F_q. For a length-m string X, H_b(X)=Σ(j=0..m−1) X[j]・b^(m−1−j); use b⁰=1 even when b=0.

For fixed distinct X,Y, D(b)=H_b(X)−H_b(Y) is a nonzero polynomial of degree at most m−1. A nonzero field polynomial has at most its degree many roots, hence collision probability ≤min(1,(m−1)/q), not unconditionally 1/q.

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Root bound and expected running time

If r is a root, factor $D(x)=(x-r)Q(x)$. Every other distinct root is a root of Q; repeating the argument bounds the number of roots by the degree.

Let $N=n-m+1$ and z be the number of true matches. Linearity of expectation over the nonmatching-window indicators needs no independence between collisions: $E[f] \leq (N-z)\min(1,(m-1)/q)$. With character-by-character verification, expected time is $O(n+m+mz+m(N-z)\min(1,(m-1)/q))$, assuming O(1) field operations. The mz verification term for true matches remains.

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One false hit and two true matches

Use digit encodings, q=11, b=10, m=2, P=12, T=122312. H_b(xy)=10x+y mod 11; the pattern hash is 1.

The five windows 12, 22, 23, 31, 12 have hashes 1, 0, 1, 9, 1. Window 23 has the same hash as P but differs; character verification rejects it. Positions 0 and 4 are true matches.

q=11, b=10: equal hashes still require verification

s = 0	s = 1	s = 2	s = 3	s = 4
12 → 1	22 → 0	23 → 1	31 → 9	12 → 1
match	skip	false hit	skip	match

P=12 hashes to 1; verify, output 0 and 4, reject 23.

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Exact collision probability in the example

Now choose b uniformly from F_{11} rather than fixing b=10. The collision equations for nonmatching windows 22, 23, 31 against P=12 have the respective solutions b=0, b=10, b=6; each collides with probability 1/11.

With collision indicators I_1, I_2, I_3, $E[f]=E[I_1]+E[I_2]+E[I_3]=3/11$. No independence is needed. At b=10, only 23 is a false hit, and both true matches still require verification.

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Automaton state

State q is the length of the longest suffix of the scanned text that is also a prefix of P, with $0 \leq q \leq m$. After $\delta(q,c)$, report end−m+1 when q=m.

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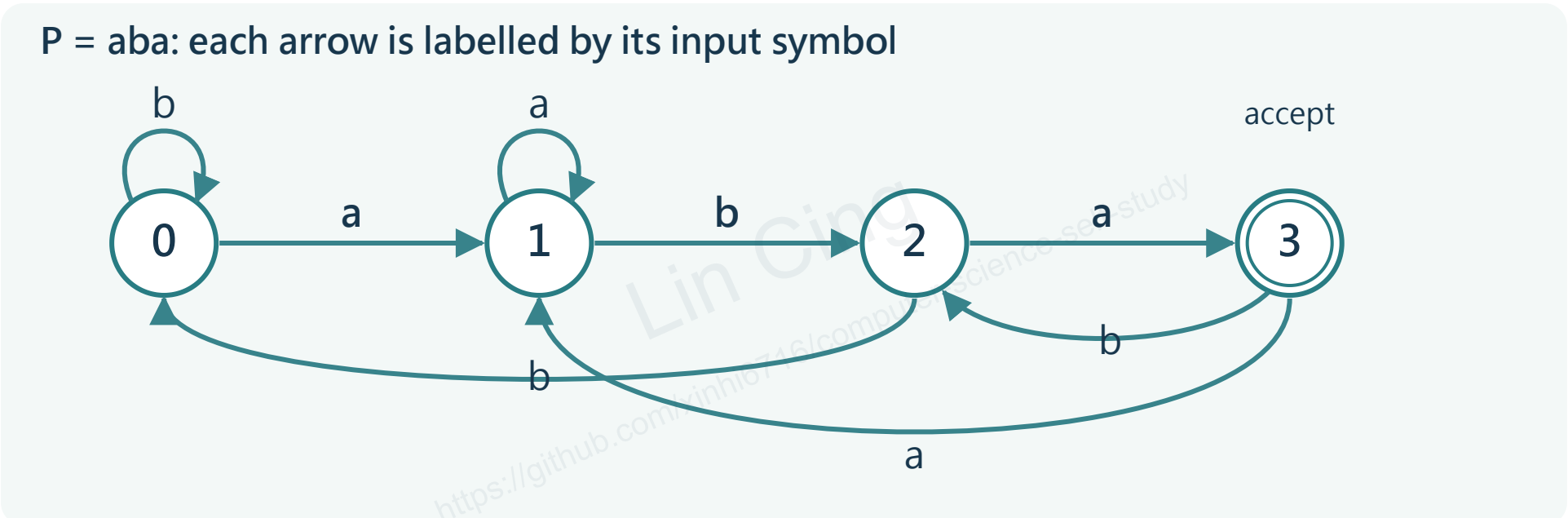
DFA transition definition

$\delta(q,c)=\max\{k \leq m: P[:k] \text{ is a suffix of } P[:q]+c\}$. Include transitions out of accepting state m so overlapping matches are found.

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DFA transition table

For P=aba and alphabet {a,b}, transitions (a,b) are 0:(1,0), 1:(1,2), 2:(3,0), 3:(1,2). Scanning takes O(n); the table uses O(m|Σ|) space.



Automaton state as a longest suffix

Let $P\langle k\rangle$ be the pattern prefix of length k, with $P\langle 0\rangle=\varepsilon$; P[i] denotes the zero-based character at index i. $\sigma(x)$ is the length of the longest pattern prefix that is a suffix of x. State $q=\sigma(x)$, and $\delta(q,a)=\sigma(P\langle q\rangle a)$.

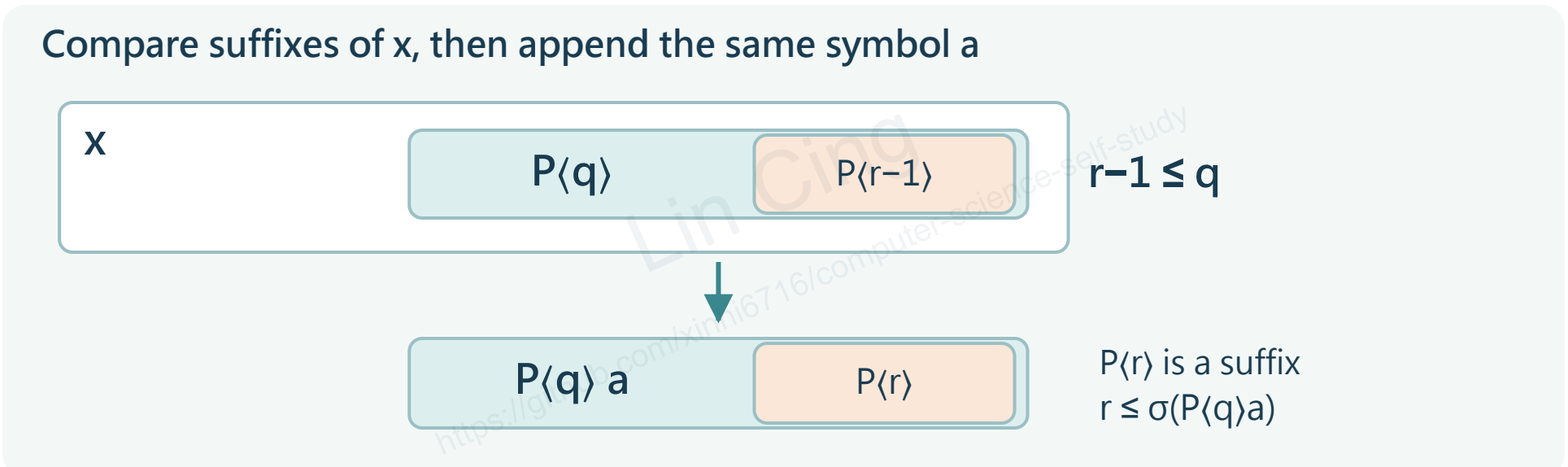
The key identity is $\delta(\sigma(x),a)=\sigma(xa)$: the state suffices to update the match without storing the entire text prefix x.

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State identity: the first inequality

Let $q=\sigma(x)$, $r=\sigma(xa)$. If $r>0$, removing the last character of $P\langle r\rangle$ shows that $P\langle r-1\rangle$ is a suffix of x, so $r-1\le q$.

Both $P\langle q\rangle$ and $P\langle r-1\rangle$ are suffixes of x; the shorter is a suffix of the longer. Thus $P\langle r-1\rangle$ is a suffix of $P\langle q\rangle$, and after appending a, $P\langle r\rangle$ is a suffix of $P\langle q\rangle a$. Hence $r\le\sigma(P\langle q\rangle a)$. For $r=0$ the inequality is immediate.



State identity: converse and correctness

Since $P\langle q\rangle$ is a suffix of x, $P\langle q\rangle a$ is a suffix of xa; every suffix of $P\langle q\rangle a$ is also a suffix of xa. Thus $\sigma(P\langle q\rangle a) \leq \sigma(xa) = r$. Together the inequalities give equality.

Initially $\sigma(\varepsilon) = 0$. Induction on the input establishes that the state is σ(the text read so far). State m occurs exactly when P is a suffix of the scanned text, which is exactly a match. Continue transitions after acceptance to detect overlap.

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Build the DFA using π

Let $\pi[q-1]$ be the length of the longest proper border of $P\langle q\rangle$. For finite alphabet Σ and $m \geq 1$:

- $q=0$: $\delta(0,a)=1$ if $a=P[0]$, and 0 otherwise.
- $1 \leq q \leq m$: if $q<m$ and $a=P[q]$, set $\delta(q,a)=q+1$; otherwise set $\delta(q,a)=\delta(\pi[q-1],a)$.

At $q=m$ do not read $P[m]$: use the second branch. Since $\pi[q-1]<q$, the required row has already been computed.

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Why border fallback builds the table

When direct extension fails, a match of length q+1 is impossible. For any shorter nonempty match $P\langle k\rangle$, dropping its last character leaves a proper border $P\langle k-1\rangle$ of $P\langle q\rangle$. All proper borders occur on the π chain, so retrying the same a from the longest proper border loses no candidate; k=0 remains possible.

Computing π uses O(m) time and space. There are m+1 rows and $|\Sigma|$ constant-time cells per row: table time and space $O(m|\Sigma|)$, assuming O(1) alphabet and table access. Scanning costs O(n) plus output.

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Prefix function

$\pi[i]$ is the length of the longest proper prefix of $P[:i+1]$ that is also its suffix, with $0 \leq \pi[i] \leq i$. Proper excludes the whole string.

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Prefix function example

For P=ababaca at indices 0..6, π=[0,0,1,2,3,0,1]. Symbols a b a b a c a correspond to 0 0 1 2 3 0 1.

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Build the prefix function token by token

i=0:a→0; 1:b differs from a→0; 2:a=a→1; 3:b=b→2; 4:a=a→3; 5:c≠b, 3→1, c≠b, 1→0, c≠a→0; 6:a=a→1.

i = 5: repeated fallback while comparing c

index	0	1	2	3	4	5	6	
	a	b	a	b	a	c	a	
k = 3			a	b	a	b ≠ c		3 → 1
k = 1					a	b ≠ c		1 → 0
k = 0						a ≠ c		π[5] = 0

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Why fallback works

If k characters match but the next does not, a shorter candidate must be a border of P[:k]. The longest candidate is π[k−1]. Repeat until a match or k=0. For P=ababaca, k=5 and next symbol x, fallback is 5→3→1→0: one fallback is insufficient.

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Why the border chain is complete

Let B be the longest proper border of X. Any shorter proper border C is both a prefix and a suffix of X, hence a prefix and a suffix of B, and therefore a border of B. Conversely, any border of B is a border of X.

Repeatedly apply π[length−1], beginning at |X|. Length strictly decreases to 0; the two containments prove that the chain lists exactly all proper border lengths, without omissions.

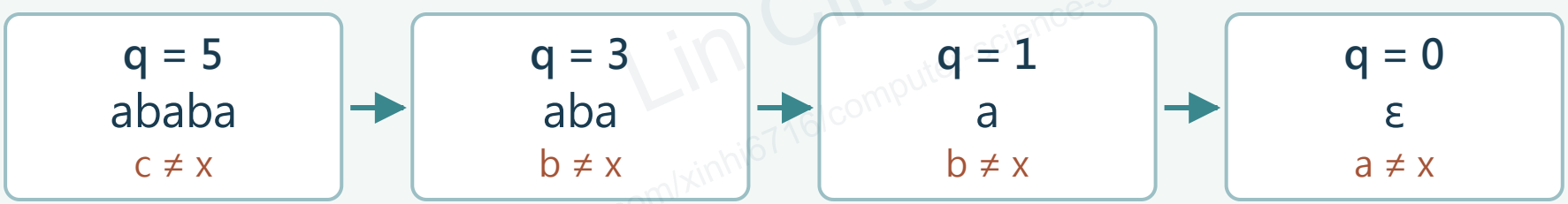
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Prefix function: invariant and proof

If m=0, π is empty. Otherwise set π[0]=0: one symbol has no nonempty proper border. For i=1,…,m−1, known earlier entries give k=π[i−1]. While k>0 and P[k]≠P[i], set k←π[k−1], keeping P[i] fixed.

Candidates are visited from longest to shortest. On equality set π[i]=k+1; if k=0 and P[0] differs, set π[i]=0. This completes the induction.

P = ababaca: retry the same symbol x



5 → 3 → 1 → 0; at q=0 compare P[0], then stay at 0.

Every shorter border is a border of the longest one.

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Why π takes linear time

Use the nonnegative potential k. Each i can increase k at most once, by at most one; each while fallback lowers k by at least one. Starting at zero, total fallbacks cannot exceed total increases $\leq m-1$.

Each i also performs at most one terminating comparison and constant work. Thus preprocessing is O(m); the same argument gives O(n) for the KMP scan. This bounds the entire run, although one input character can trigger multiple fallbacks.

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KMP preprocessing

Set $\pi[0]=0$. For $i=1..m-1$, start $k=\pi[i-1]$; while $k>0$ and characters differ set $k=\pi[k-1]$; increment k on equality; set $\pi[i]=k$.

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KMP scan

Start $q=0$. For each c, while $q>0$ and $P[q] \neq c$ set $q=\pi[q-1]$. Increment q on equality. At $q=m$ report $i-m+1$, then set $q=\pi[m-1]$ to preserve overlaps.

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Two KMP scan states

Assume $m \geq 1$. Let $\sigma(x)$ be the longest pattern-prefix suffix of x, allowing length m; let $\rho(x)$ be the longest such prefix with length strictly less than m.

At the start of each iteration $q=\rho(x)$, where x is the text read so far; initially $q=\rho(\varepsilon)=0$. If $\sigma(x)<m$, then $\rho(x)=\sigma(x)$. If $\sigma(x)=m$, then $\rho(x)=\pi[m-1]$. Thus the stored post-match state is the longest proper border, not m.

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A scan step simulates the DFA

For the same new symbol a, the while loop tries the border chain from q in decreasing order. The first extendable candidate is longest; if none extends, the state is 0. Thus the temporary state $r=\delta(q,a)$.

If $\sigma(x)<m$, $q=\sigma(x)$; if $\sigma(x)=m$, $q=\pi[m-1]$ and $\delta(m,a)=\delta(\pi[m-1],a)$. By $\delta(\sigma(x),a)=\sigma(xa)$, in both cases $r=\sigma(xa)$.

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Why post-match fallback preserves overlaps

r=m exactly when the pattern is a suffix of the scanned text xa, so output i−m+1 is correct. Then q←π[m−1]=ρ(xa). If r<m, set q←r=ρ(xa). Either way the next iteration's invariant is restored.

For P=aba and T=ababa, positions 2 and 4 produce outputs 0 and 2. After either acceptance, retain q=1; extending that border preserves the overlapping occurrence.

P=aba, T=ababa: accept first, retain the border

i	0	1	2	3	4
T[i]	a	b	a	b	a
q before	0	1	2	1	2
r = σ	1	2	3	2	3
q after	1	2	1	2	1
output	–	–	0	–	2

When r=3 report, then set q=π[2]=1.

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KMP token trace

T=ababa, P=aba, π=0,0,1. Each i, symbol, q: 0 a→1; 1 b→2; 2 a→3, report 0 and fall back to 1; 3 b→2; 4 a→3, report 2 and fall back to 1.

P = aba: retain border length 1 after a match

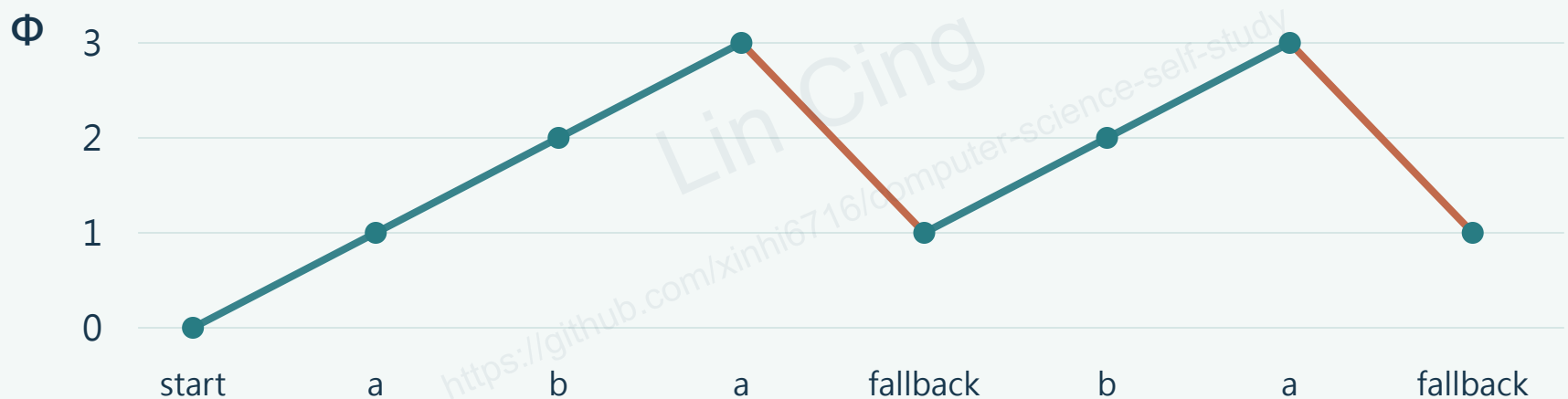
T index	0	1	2	3	4
input	a	b	a	b	a
state q	1	2	3	2	3
match at			0 3 → 1		2 3 → 1

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Amortized linear time

Use q or k as potential in each preprocessing or scan pass. Each input position increases it by at most 1; every nonzero fallback decreases it by at least 1. Total fallbacks cannot exceed total increases. Other comparisons cost O(1) per position, so total time is O(m+n).

Potential Φ = q: each input raises it at most once



Every fallback drops Φ by at least 1; total drops ≤ total rises.

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KMP time and memory

Preprocessing costs O(m), scanning O(n), and the π table O(m). Storing z outputs costs an additional O(z); streaming output avoids that storage.

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Four methods compared

For $1 \le m \le n$, naive takes worst-case $O(nm)$ time and $O(1)$ space; Rabin–Karp worst-case $O(nm)$ and $O(1)$ workspace; DFA $O(n)$ scan and $O(m|\Sigma|)$ table; KMP $O(n+m)$ and $O(m)$.

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Implementation pitfalls

Specify the symbol and index unit: bytes, Unicode code points, and grapheme clusters differ. Handle the empty pattern first; verify hash hits; do not reset to zero after a match.

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Verification strategy

Use direct enumeration as an oracle and exhaust all short texts and patterns over a small alphabet. Compare complete outputs of Rabin–Karp, DFA and KMP, including collisions, empty patterns, m>n, and overlaps.

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Takeaway

Borders of the matched prefix form the viable fallback chain. KMP follows it on demand; the DFA precomputes state updates; hashing requires verification to guarantee correctness.

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References and attribution

CLRS, Introduction to Algorithms, 3rd ed., chapter 32, pp. 990–1006. MIT 6.046J Fall 2004, Lecture 22.

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